
Paz Hameiri

**CON trolling the
weather**

About myself

- I'm a system engineer
- M.Sc. in Electro-Optical Engineering
- Worked fourteen years in the aerospace industry
- Six years of experience with telecommunication systems design
- Launched two weather balloons with elementary school pupils
- DEF CON 29 speaker
- BSidesTLV 2023 speaker



Why trolling the weather???

- I read an article about a country, that's trying to launch a satellite into space despite a string of failures
- It reminded me of a sentence from an old James Bond book:
“Once is happenstance.
Twice is coincidence.
Three times is enemy action”

— Ian Fleming, Goldfinger

Why trolling the weather???

- While I was in my James Bond mood, I wondered:
 - Can I stop a satellite launch?
 - Can I cause it to fail after liftoff?
 - Can I do it on my own?
 - After a couple of hours, I realized that the answer is: Yes!
And it's not that hard – I just need to control the weather.
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What's with air vehicles and the weather?

- We're all familiar with the “launch delayed due to weather conditions” headlines.
 - There is no “all-weather” air vehicle:
 - Lightning might cause electrical failures and even structural damage
 - Hail impact might cause structural damage
 - Rain and moisture might freeze on the wings and control surfaces
 - High winds might cause control problems, even on a dry day with no rain or thunderstorms nearby
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Weather balloon's introduction

- Airman 1st Class Ian Dudley, 30th Space Wing Public Affairs, Vandenberg space force base, California:
 - The balloons carry instrumentation that provides wind speed and direction, atmospheric temperature, and humidity every 100 feet from the surface of the earth to near the edge of space where the balloon finally bursts
- The instrumentation is called “Radiosonde”



Weather balloons launch sites

- Between 900 and 1,300 locations around the globe do routine releases
- Two or four launches each day



Weather balloons use

- **William Shmeiser, 30th Operations Support Squadron weather systems director and senior meteorologist, Vandenberg space force base, California:**
 - **“The data goes directly into the NOAA database and is included in the national weather models and analysis you see on television”**
 - **“Depending upon the mission we release from five to 15 balloons during a launch count, sometimes more if weather conditions demand”**
 - **“Upper air balloon support is critical to Vandenberg's launch mission as well as being a huge part of daily weather forecasting; not only on Vandenberg but across the community and the nation.”**
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NASA's criteria for safe launch

- **NASA's Falcon 9 Crew Dragon Launch Weather Criteria:**
 - Do not launch through upper-level conditions containing wind shear that could lead to control problems for the launch vehicle
 - Do not launch if downrange weather indicates violation of limits at splashdown in case of Dragon launch escape

Downrange weather is monitored at more than 50 locations along the ascent track along the North American eastern seaboard and across the North Atlantic
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Who's who in the radiosonde business?

- Top Radiosonde types, according to World Meteorological Organization launch site data:

Radiosonde type	Launch Sites	Precentage
Vaisala RS41	186	29.8
GTS-1 (China)	82	13.1
Vaisala RS92	67	10.7
Lockheed Martin LMS06	66	10.6
Meteomodem M10	48	7.7
Graw DFM-09	40	6.4
Meisei iMS-100	36	5.8

Vaisala RS41 transmission frame

- Most common models: RS-41SG and RS41-SGP
 - Basic frame: 320 Bytes, divided into blocks
 - Data whitening: XOR mask is used to generate frequent level changes
 - Byte-wise little endian: LSByte is transmitted first
 - Bitwise little endian: LSBit is transmitted first
 - Two layers of error detection and correction:
 - Each numbered block comprises CRC-16 bytes
 - Reed Solomon is applied at the complete message
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Vaisala RS41 frame

Bytes	Block	Content
8	Header	Header
48	ECC	Reed Solomon ECC
1	Frame type	Frame type
44	0x79	Status: frame number, serial number, battery voltage, status bits, sonde type, board temp., control data, subframe (0-51)
46	0x7A	Measurements: ambient & sensors temp., humidity, ambient pressure
34	0x7C	GPS info: time, space vehicles number & reception quality
93	0x7D	GPS raw data: space vehicles relative distance and velocity
25	0x7B	GPS position: ECEF position X/Y/Z, ECEF velocity X/Y/Z, number of SVs, velocity accuracy, PDOP
21	0x76	Empty block

Vaisala RS41 subframes: 51 * 16 Bytes

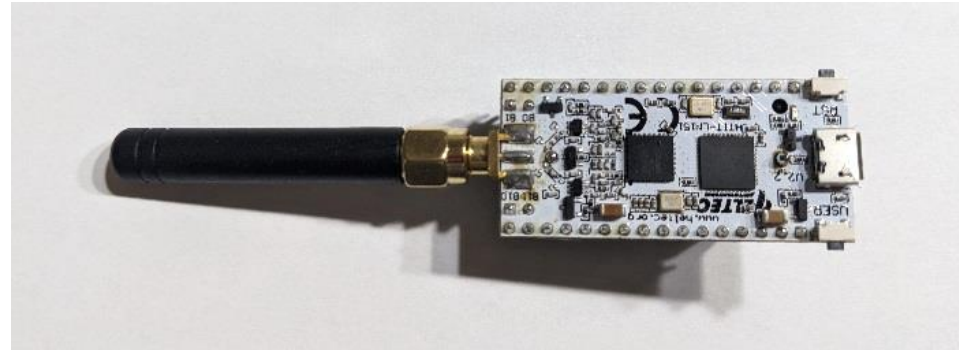
- Calibration data
 - Transmission frequency
 - Sonde type
 - Software version
 - Mainboard type
 - Components serial numbers
 - Burstkill data
 - Diagnostics data
 - More...
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Vaisala RS41 RF properties

- Frequency: 400.15 – 406 MHz
 - Output power: min 60mW (18dBm)
 - Deviation peak-to-peak: 4.8kHz
 - Modulation: GFSK
 - Data downlink: 4,800 bits/s
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Strategy 1: Jamming attack

- Use a powerful transmitter to transmit jamming signals
- My choice: HELTEC AUTOMATION Lora Node 151, 433MHz
 - SX1278 LoRa chip
 - STM32L151CCU6 MCU
 - Maximum transmission power: 20dBm
 - 1/2AA Lithium battery
 - 20\$ on eBay
 - 25 grams with the battery



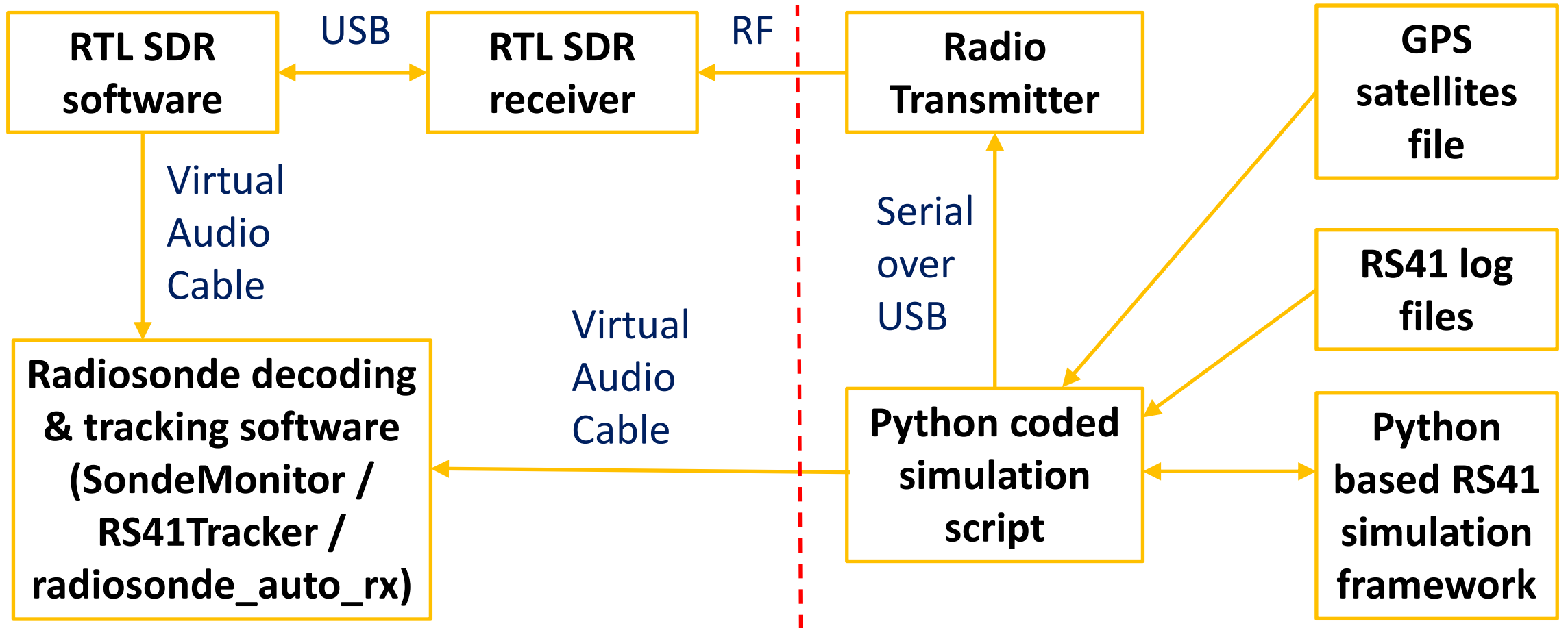
Jamming demo

1st Video clip: jamming process and results

RS41 messages decoding

- A dedicated radiosonde tracking community had done wonderful work developing tools to receive, decode, process, and store the information.
The public can access the data free of charge.
 - Radiosonde community tools: radiosonde_auto_rx, RS41Tracker, SondeMonitor, SondeHub.org, radiosondy.info
 - Credit to the radiosonde community developers: bazjo, rs1729, Zilog80, DF9DQ, IW1GIS, Mark Jessop, Michael Wheeler, COAA and others...
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RS41 simulator development environment



RS41 simulation framework

- **Functions libraries:**
 - **RS41BlockRW.py: Frame level/Block level read/write operations**
 - **RS41SubframeRW.py: Subframes level read/write operations**
 - **RS41Functions.py: GPS data calculations, measurements calculation, message level data whitening**
 - **RS41SimFunctions.py: Log file operations, radio messages generation, audio messages generation**
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Messages generation with the framework

- Synthesize frames and subframes
 - Requires a behavioral model for each measurement
 - Shortcut: Take the calibration data out of a log file
 - Use log files:
 - Alter specific data items, as required
 - Synthesize the GPS data, serial numbers, fine details, etc.
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Strategy 2: Spoofing attack

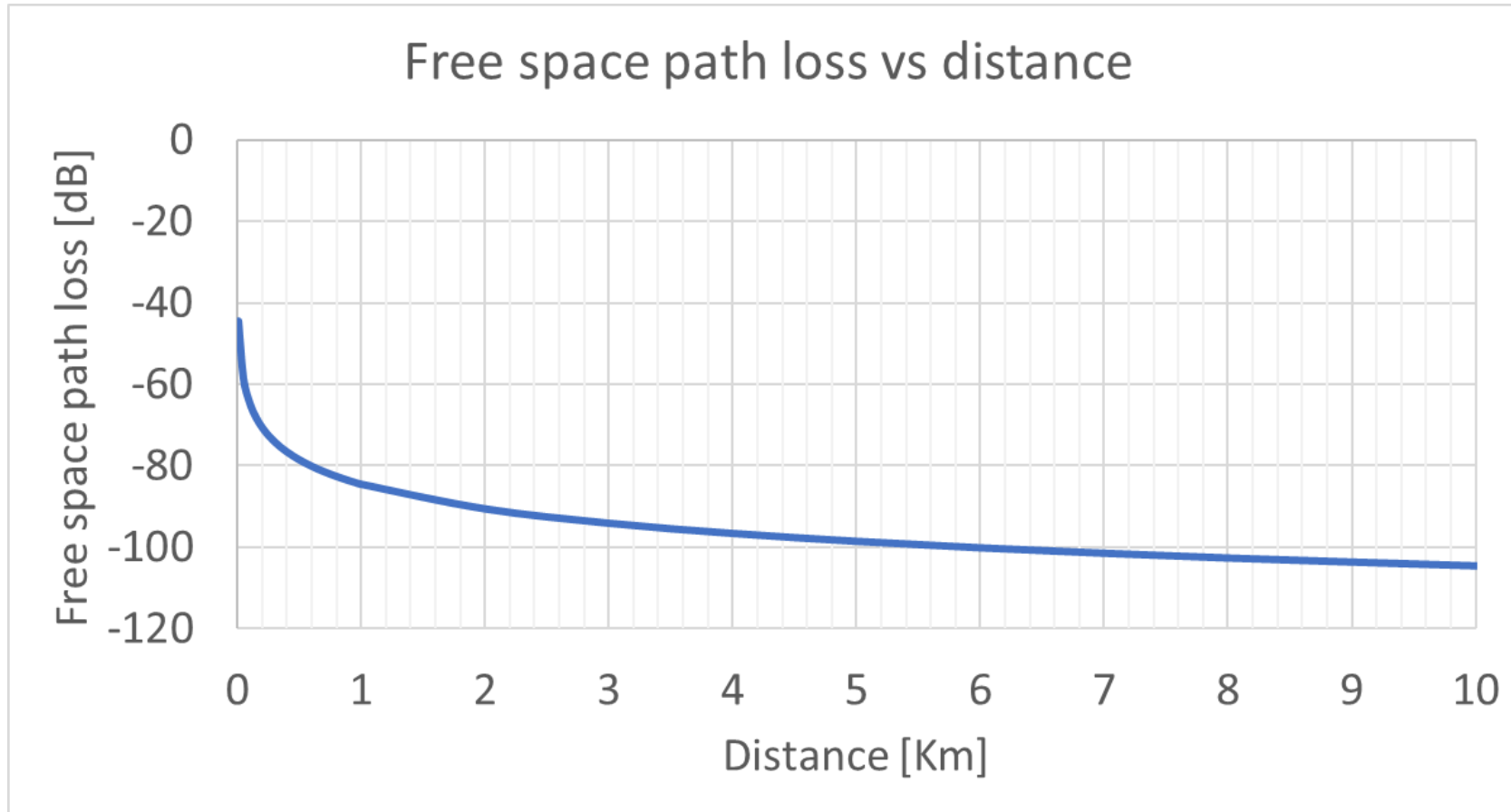
- Use a powerful transmitter to transmit spoofed messages
 - Technique:
 - Receive all the subframes from an active radiosonde (at least 51 frames)
 - Prepare for spoofing (setup the transmitter, encode message/s, etc.)
 - Jam the radiosonde transmission to raise the data's uncertainty level
 - Transmit spoofed messages
 - Jam the radiosonde transmission to raise the data's uncertainty level
 - Stop
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Ride on the signal to noise ratio

- Use a powerful transmitter to transmit spoofed messages
- At the receiver:
 - Spoofing transmitter = Signal
 - Radiosonde signal = Noise
- Requires good signal to noise ratio:

SNR [dB]	Reception
5 to 10	Indistinguishable
10 to 15	Unreliable
15 to 25	Poor
25 to 40	Good
41 or higher	Excellent

Ride on the signal to noise ratio



Ride on the signal to noise ratio

- The spoofing signal to noise ratio can be increased by using:
 - A directional antenna
 - A power amplifier
 - Elevated transmitter (Balloon/Kite/Drone/others)
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Spooofing demo

**2nd Video clip: an example of a real-time
spooofing attack**

Script walkthrough: Active flight simulation

Simulate an active flight with data from a log file:

- Define a new radiosonde ID, mainboard s/n, and pressure sensor s/n
 - Define a position offset (shift the weather balloon's position)
 - Read an updated GPS satellites file from <http://celestrak.org/>
 - Read an RS41 log file and retrieve the subframes
 - Setup the RF transmitter port/audio stream
 - Initiate the Transmission thread function: Transmit data upon event
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Script walkthrough: Active flight simulation

- Initiate the transmission loop:
 - Load a logged message
 - Recover the data with Reed-Solomon FEC
 - Update the radiosonde ID, mainboard s/n, and pressure sensor s/n
 - Calculate and update a new CRC code for the STATUS block
 - Extract the GPSPOS Block data
 - Offset the weather balloon's position
 - Calculate and build new GPSPOS, GPSRAW, and GPSINFO blocks
 - Prepare the data for transmission (RF transmitter/Audio stream)
 - Wait for a time event and trigger the transmission thread function
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A simple spoofing attack script demo

3rd Video clip: the results of the simple spoofing attack

RS41-SGM - Introduction

- Intended for military use
 - Radio silence until a specified height or time has been reached
 - “Radiosonde transmission does not reveal the balloon launch location”
 - The data is recorded during the radio silence
 - “Transmitted data is also encrypted”, 128-bit key
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RS41-SGM – Military? Are you serious?

- Most properties were not changed:
 - Transmits in known, monitored, civilian frequency band
 - Transmits with known properties: modulation, bit rate, etc.
 - Transmits a known frame structure with known data items: Frame number, serial number, etc.
 - Only the measurements and the GPS data are encrypted
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RS41-SGM – Military? Are you serious?

- The encryption is not always activated
 - Why?
 - When the encryption is not activated:
 - The RS41-SGM use the RS41-SG's framing structure
 - The RS41-SGM identifies as an RS41-SGM
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RS41-SGM – Military? Are you serious?

- “Radiosonde transmission does not reveal the balloon launch location”?
 - There are numerous positioning techniques for a constantly transmitting object in the line of sight
Latest state-of-art crowd-funded project: KrakenSDR
 - Once the position is known, it's not hard to trace back to the launch site using wind predictions (similar to landing predictions)
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RS41-SGM – Military? Are you serious?

- William Shmeiser, Vandenberg space force base, California:
“Depending upon the mission we release from five to 15 balloons during a launch count, sometimes more if weather conditions demand”
 - It means that frequent RS41-SGM launches indicate that the military is up to something...
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Coping with spoofing

- **Don't encrypt the messages**
 - Keys management is a headache
 - Synchronizing reception sites is a headache
 - Requires significant upgrades
 - Might not be welcomed by some customers
 - Bad for the radiosonde community
 - Information collection is public-funded – the data belongs to the public
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Coping with spoofing

- **Authenticate the messages**
 - **Modify the messages to comprise an authentication tag**
 - **Customers who are not interested in message authentication can keep using their equipment as is**
 - **The community can keep using their equipment as is**
 - **Authentication can be done offline**
 - **A potential business model:**
 - “Want to check the authenticity of your data???**
 - Upload your log file to our website and get an immediate answer!!!**
 - Only 9.95\$!!!”**
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Epilogue

- The framework should be used to develop and test tools for the radiosonde community
- Source code:
 - <https://github.com/CuriosityRocks/RS41Simulator>
- Thank you 😊



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